

After the Acquisition

Post-Merger Innovation Among Stayers in Pharma M&A

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Regulation, Markets and Environment Research Seminar
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April 2026

Why this matters

- High concentration, frequent M&A, and exceptionally high R&D intensity make pharma the central industry for studying merger-innovation links
- FTC 2023 *Multilateral Pharmaceutical Merger Task Force*: “What is the full range of a pharma merger’s effects on innovation?”

What the literature establishes

- Patenting falls 10–30% post-merger at the firm level (Ornaghi 2009; Haucap et al. 2019)
- Exit and departure rates spike; exiters and movers each lose roughly two patents, around 30% of expected career output (Cassi & Ornaghi 2026)
- **Open gap:** productivity, quality, and field trajectories of inventors who *stay* remain unexplored

Research Question

Do acquisitions reduce the patent quantity, citation-weighted quality, and technological focus of target-firm inventors who remain with the acquiring group, relative to comparable inventors not yet exposed to a deal?

Theoretical mechanisms

- **Integration frictions** force researchers to navigate new processes at the expense of discovery (Kapoor & Lim 2007)
- **Loss of status** reduces motivation for inventors central to the pre-merger R&D structure (Paruchuri et al. 2006)
- **Disrupted knowledge networks** limit access to tacit knowledge (Ernst & Vitt 2000)
- **Weaker championing** of internal ventures as focus shifts to existing product lines (Hitt et al. 1991)

Hypotheses

- H1** Acquisition reduces post-merger patent *quantity*
- H2** Acquisition reduces citation-weighted *quality*
- H3** Stayers exhibit *TechDrift*, shifting into the acquirer's technological fields

Heterogeneity predictions

Interaction	Predicted sign
Age $\times$ After	+
Exclusivity $\times$ After	-
CommonIPC $\times$ After	+
Big deal $\times$ After	-

Evaluation and integration frictions drive the cross-sectional patterns.

Base data — Cassi & Ornaghi (2026, EER)

- EPO/PatStat (v. 2017) and Orbis/Zephyr (Bureau van Dijk)
- 311,358 inventors; 1,000,751 inventor $\times$ year observations, 1988–2015
- 513 pharma acquisitions; avg. 3.21 observations per inventor
- Upstream cleaning taken as given: disambiguation, original-applicant recovery, dynamic group construction

Own additions

- DuckDB/R pipeline producing `inventor_year`, `inventor_ipc_year`, `group_ipc_year`
- OECD Patent Quality Indicators (PQII) linked via application ID: originality, radicalness, claims, composite score
- *TechDrift* (in progress): IPC cosine similarity to pre-merger baseline

$$\text{TechDrift}_{it} = \frac{\mathbf{v}_i^t \cdot \mathbf{v}_i^0}{\|\mathbf{v}_i^t\| \|\mathbf{v}_i^0\|}$$

Declining values signal movement into unfamiliar fields.

Sample construction

- *Treated*: inventors filing ≥ 1 patent for the target in the 5 pre-acquisition years who continue filing for the acquiring group within $x \in \{3, 5\}$ years; acquirer-side inventors excluded
- *Control*: not-yet-treated inventors whose employer will be acquired in a later cohort, more comparable than never-treated firms which differ structurally

Estimator

- Callaway & Sant’Anna (2021) group-time ATT avoids TWFE contamination under staggered timing and heterogeneous effects
- Doubly robust estimation; covariates: PrePatents, Age, Tenure, Exclusivity, FirmPatentStock

Estimating equations

Event study (dynamic ATT):

$$Y_{ict} = \sum_{\tau=-5}^{+5} \lambda_{\tau} \cdot \mathbf{1}[t - g_c = \tau] + \alpha_i + \xi_{ct} + \varepsilon_{ict}$$

α_i inventor fixed effects ξ_{ct} firm $\times$ year fixed effects
Pre-treatment coefficients test parallel trends (Roth 2022).

Aggregate ATT:

$$Y_{ict} = \lambda \cdot \text{After}_{ict} + \alpha_i + \xi_{ct} + \varepsilon_{ict}$$

$\lambda < 0$ indicates a post-acquisition decline in output.

Novel contributions

- Prior inventor-level M&A studies measure only raw counts; OECD PQII provides the first quality-adjusted picture of stayer performance
- TechDrift captures field-level misallocation post-merger; the application to stayer career trajectories is new
- CS (2021) staggered DiD with heterogeneity analysis and selection correction at inventor level

Expected results

- Patent quantity and quality decline post-acquisition
- TechDrift rises as stayers shift into the acquirer's fields
- Non-collaborating stayers are substantially more affected

Policy implication

Stayer effects add an additional channel of dynamic welfare loss to pharma merger review. With \$180bn in patent revenues expiring in 2027–28, pharma's acquisition-driven pipeline strategy carries an underappreciated cost: if acquisitions suppress the innovation output of retained inventors, they risk deepening the very pipeline gap they are meant to close.